

The Full Generic Binary Tower of $X^2 + 1$: New Branch Inertia at Every Height

HCS-C393 theorem and reproducibility package

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Abstract

The polynomial $X^2 + 1$ generates the full automorphism group of its generic binary preimage tree at every height. A new critical value produces one sibling transposition; conjugation and restriction then force every last-level swap, without relying on a finite Galois computation. Exact small-tree permutations, critical-value identities and residue-field computations audit the normalization and its failure boundaries. These are source-local reconstructions with classical Galois and periodic-density owners. A transcendental generic basepoint, a Frobenius cycle and an original-map periodic orbit are not interchangeable, and no target prime-period or spectral identity is asserted.

Keywords: arboreal Galois group; quadratic dynamics; ramification; wreath product; cycle index; periodic density.

中文摘要

本文对二次映射的泛型逆像树证明每一层 Galois 群都是完整二叉树自同构群。每层新增临界值给出一个兄弟交换，再由共轭产生全部末层交换。文中区分泛型基点、数值特化、原映射时间和 Frobenius 时间，经典理论明确归属，不把算术源结构写成目标零点对应。

关键词：树状Galois群；二次动力学；分歧；圈积；循环指标；周期点密度。

1 An intrinsic arithmetic source, not a fitted prime clock

Let $f(X) = X^2 + 1$, let t be transcendental over $\mathbb{Q}$, and let K_n be the splitting field of $f^n(X) - t$ over $\mathbb{Q}(t)$. The rooted tree T_n has root t , level- j vertices $f^j(X) = t$, and edges $x \mapsto f(x)$. Write

$$G_n = \text{Aut}(T_n), \quad d_n = |G_n| = 2^{2^n - 1}.$$

The generic basepoint is part of the frozen source. No theorem below says that every specialization $t = a \in \mathbb{Q}$ has this Galois group.

Odoni [1] established the Galois theory of iterated polynomials. Pink [2] studies quadratic iterated monodromy with infinite postcritical orbits; not every quadratic rational map in that broader class has a full tree group. We give the specific new-branch proof needed for this polynomial. The wreath-product approach to periodic proportions belongs to Juul–Kurlberg–Madhu–Tucker [3]. Our finite audit and explicit source comparison do not certify global literature novelty.

Within the project, C374 treated the numerical basepoint-two power-map tower, with an affine index-two image. The present source has full nonabelian wreath growth, new ramification at every height and an original finite-field dynamical consequence. It is not another parameter table for that affine tower. All claims below concern native arithmetic; they do not insert a prime list or a logarithmic roof into the map.

2 The complete group from one new branch

Set $c_0 = 0$ and $c_i = f^i(0)$, so that

$$0, 1, 2, 5, 26, \dots$$

are distinct: $c_{i+1} = c_i^2 + 1 > c_i$ for $i \geq 1$. The chain rule gives

$$(f^n)'(X) = 2^n \prod_{j=0}^{n-1} f^j(X). \quad (1)$$

Thus the finite branch values at height n are precisely $c_1, \dots, c_n$. The rational function f^n has degree 2^n , so $[\overline{\mathbb{Q}}(X) : \overline{\mathbb{Q}}(f^n(X))] = 2^n$. Equivalently $f^n(X) - t$ is irreducible over $\overline{\mathbb{Q}}(t)$, and geometric monodromy is transitive on its leaves.

Theorem 1. *For every $n \geq 1$, both the geometric and arithmetic Galois groups of $f^n(X) - t$ are G_n . The restriction to height $n - 1$ is onto with kernel $(C_2)^{2^{n-1}}$. The infinite tower has group $\text{Aut}(T_\infty)$.*

Proof. At $t = c_n$ the only critical point in the fibre is $X = 0$. Indeed $f^j(X) = 0$ for $j \geq 1$ would imply $f^n(X) = c_{n-j} \neq c_n$. Near zero,

$$f^n(X) = c_n + A_n X^2 + O(X^4), \quad A_n = 2^{n-1} \prod_{i=1}^{n-1} c_i \neq 0.$$

Local inertia is therefore a single transposition of two leaves. They are siblings because their common image under f is 1. The previous-height cover is unramified at this new branch value, so the transposition belongs to the last-level restriction kernel.

Inductively the group at height $n - 1$ is G_{n-1} ; restriction between splitting fields is surjective. The new kernel is contained in the group of independent swaps of the 2^{n-1} sibling pairs. Conjugate the one new swap by elements transitive on their parents. Every sibling swap is obtained, and these generate the entire kernel. The degree-two case starts the induction.

The arithmetic group contains the full geometric group and embeds in G_n , so it too is full. Thus the constant field is $\mathbb{Q}$. Compatible surjective restrictions give the full inverse limit: an automorphism of the union is exactly a family of compatible finite-level automorphisms. $\square$

This proof derives the kernel rather than inferring it from a group-order fit. It also records why a post-critical collision would break the argument.

3 Conclusion

New critical inertia closes the entire generic binary Galois tower. This is one paper, not three publications; its present record is Round zero. The accompanying proof ledger, independent checks and source audit delimit the complete source theorem from the unachieved target bridge.

References

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